

Name: _____

R. Hammack

Score: _____

Directions: Put away phones, calculators, smart watches, etc. Answer in the space provided.

1. (4 points) Consider the vectors $\mathbf{u} = (8, -2, 0, 1)$, $\mathbf{v} = (3, -2, 1, 1)$ and $\mathbf{w} = (1, 4, 4, 1)$. in $\mathbb{R}^4$. Find $\mathbf{x}$, given that $\mathbf{w} - 2\mathbf{x} = \mathbf{u} - 3\mathbf{v}$.

2. (8 points) Say if each of the following sets is a vector space. (If so, just answer “Yes.” If it is **not** a vector space, then answer “No” and state a vector space property that fails.)

(a) $\{(x, y) \in \mathbb{R}^2 : x - y = 0\}$

(b) $\left\{ \begin{bmatrix} x & x+1 \\ x & x \end{bmatrix} : x \in \mathbb{R} \right\}$

(c) $\left\{ \begin{bmatrix} 0 & x \\ 2x & 0 \end{bmatrix} : x \in \mathbb{R} \right\}$

(d) P_9

(e) $M_{9,3}$

(f) $\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$

(g) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > 0\}$

(h) $\{(x, y, z) \in \mathbb{R}^3 : x = 0\}$